

Nabankur Ray

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🌐 https://analyticswithnaban.com/
📁 https://nabankur14.github.io/

SKILLS

Programming Languages

Python (Advanced)
R
SQL
PySpark
JavaScript (Basic)

Frameworks & Libraries

Scikit-learn
XGBoost
LightGBM
Keras/TensorFlow
Random Forest
SVM
Ensemble Methods
NLP & Gen AI
Transformer Architecture
LLM Fine-tuning
Prompt Engineering
Azure OpenAI (GPT)

Cloud Platforms

Azure Cognitive Services
Azure ML
Azure Bot Service
Azure AI Search
Azure App Service

BI & Viz

Tableau
Power BI (DAX, Power Query)
Matplotlib
Seaborn
Plotly

Data Eng.

Pandas
NumPy
PySpark
XML Parsing
REST APIs
Streamlit

Methodologies

Data Scientist and ML Engineer combining a manufacturing operations background with advanced analytics expertise. Currently pursuing a Master's in Data Science at Deakin University (Rank 1, Trimester 3) alongside a PGP in Data Science & Business Analytics from UT Austin & Great Lakes (GPA: 3.86/5). Builds and deploys end-to-end ML, NLP, and cloud AI solutions across FinTech, Healthcare, and Retail.

PROFESSIONAL EXPERIENCE

Production Planning & Control Engineer

Delta Electronics India Pvt. Ltd. • Hosur, Tamil Nadu •
Jul 2021 - Jun 2023

- Led a team of four to execute a full factory transfer project, maintaining 100% on-time delivery for Solar Inverters, VFD Drives, and Power Supplies.
- Managed production planning in SAP, improving efficiency by 12% and reducing dispatch delays by 15%.
- Coordinated BOM changes, MRP/MPS planning, and inventory monitoring to ensure production accuracy.
- Managed international purchasing and multiple product lines, maintaining on-time deliveries for 23 months.

Data Analytics & Visualization Virtual Internship

Accenture North America (via Forage Job Simulation) • null •
Oct 2024 - Nov 2024

- Cleaned, modeled, and analyzed multiple social media datasets to uncover content performance insights for a simulated client engagement.
- Designed stakeholder-facing dashboards and client presentations, translating data-driven findings into strategic business recommendations.

PROJECTS

ShopSmartBot — Multimodal AI E-Commerce Recommendation Bot

https://github.com/nabankur14/shopsmart-multimodal-recommendation-bot
• Apr 2026 - Apr 2026

Azure OpenAI GPT Azure Bot Framework SDK v4 RAG
Azure AI Search Python aihttp

- Architected a five-layer multimodal AI bot (text, voice, image) achieving sub-3s response latency and SUS usability score of 81.5.
- Implemented a RAG pipeline with hybrid keyword and vector retrieval, grounding all LLM responses in live product catalogue data.

Azure CV Object Detection & Tracking

https://github.com/nabankur14/azure-cv-object-detection-tracking •
Apr 2026 - Apr 2026

Azure Computer Vision API Python OpenCV PASCAL VOC
IoU Tracking

- Built an end-to-end cloud AI pipeline achieving F1 = 1.000 and Mean IoU 0.73–0.84 across object classes on real-world urban street scenes.
- Engineered an IoU-based greedy tracking algorithm across 192 video frames (Track F1 = 0.821), reducing API consumption by 80% via frame subsampling.

PySpark Business Review Analytics (Score: 88/100)

EDA
A/B Testing
Hypothesis Testing
Time Series (ARIMA/SARIMAX)
Clustering
RAG

Tools & Technologies

Git/GitHub
SAP
Google Colab
Jupyter
KNIME
VS Code

ACHIEVEMENTS

Rank 1, Trimester 3

Master of Data Science (Global), Deakin University (2026)
• Feb 2026 - Jul 2026

Ranked 1st among peers in Trimester 3.
Recognized by Great Learning for excellent academic performance

CERTIFICATES

Generative AI Certification

Great Learning • Dec 2025 - Dec 2026

Mastering Big Data Analytics

Great Learning • Dec 2025 - Dec 2026

Data Visualization using Power BI

Great Learning • 2025 - 2026

Data Analytics & Visualization Simulation

Accenture North America (Forage) • Oct 2024 - Nov 2024

<https://github.com/nabankur14/google-maps-reviews-analytics-pyspark>
• Oct 2025 - Oct 2025

[PySpark](#) [NLP](#) [ALS Collaborative Filtering](#) [ARIMA](#) [CountVectorizer](#)

- Built a scalable analytics pipeline on 500K+ Google Maps reviews; ALS recommendation system achieved RMSE of 0.9896.
- Applied NLP (CountVectorizer, WordCloud) to identify top 30 keywords and segment low-rating reviews for dissatisfaction root-cause analysis.
- Implemented ARIMA(1,0,1) time series forecasting with grid search across 27 parameter combinations, revealing COVID-19 impact on review activity.

BCCI Cricket Win Prediction — Capstone (Score: 94/100)

<https://github.com/nabankur14/cricket-win-prediction-bcci-ml> • May 2025 - May 2025

[Python](#) [Scikit-learn](#) [GridSearchCV](#) [Logistic Regression](#)

[Decision Tree](#) [KNN](#)

- Developed four supervised ML classifiers on 2,930 records; tuned Decision Tree achieved 95.5% recall and 0.927 F1-score.
- Applied VIF-based multicollinearity removal, p-value elimination, and ROC threshold optimization (0.847) for production-grade performance.
- Delivered five opponent-specific match-winning strategies for BCCI using feature importances across three upcoming series.

Time Series Wine Sales Forecasting (Score: 60/60)

<https://github.com/nabankur14/time-series-wine-sales-forecasting> • Jan 2025 - Jan 2025

[Python](#) [ARIMA](#) [SARIMAX](#) [Exponential Smoothing](#) [statsmodels](#)

[pmdarima](#)

- Engineered an end-to-end forecasting pipeline on 15 years of ABC Estate Wines sales data (Rose & Sparkling variants).
- Conducted seasonal decomposition, ADF stationarity tests, and RMSE-based model selection across ARIMA, SARIMAX, and ETS models.
- Delivered executive recommendations on peak-season inventory planning and targeted marketing ROI optimization.

EDUCATION

Master of Data Science (Global)

Data Science • Deakin University, Australia • Jul 2025 - Present • null

Post Graduate Program

Data Science & Business Analytics • University of Texas at Austin & Great Lakes Institute of Management (Dual Degree)
• May 2024 - May 2025 • 3.86 / 5.0

Bachelor of Engineering

Electrical & Electronics Engineering • Thiagarajar College of Engineering, Madurai • Jul 2017 - May 2021 • 9.1 / 10